Simulation-Based Inference

Exercise 3

Predicting article production by students in biochemistry Ph.D. programs

Use the `bioChemists.csv` dataset which contains a sample of 915 biochemistry Ph.D. students. We will focus on the following variables:

- `art`: count of articles produced during last 3 years of Ph.D.
- `phd`: prestige of Ph.D. department
- `ment`: count of articles produced by Ph.D. mentor during last 3 years

Start with a Poisson regression model, using the prestige of the Ph.D. department and production of the Ph.D. mentor to predict the production of the Ph.D. student,

$$\begin{aligned} \text{art}_i &\sim \text{Poisson}(\lambda_i) \\ \lambda_i &= \exp(b_0 + b_1 \text{phd}_i + b_2 \text{ment}_i) \end{aligned}$$

(a) Use a simulation-based test for the hypothesis $H_0 : b_1 = 0$ vs. $H_1 : b_1 \neq 0$.

The Poisson regression model assumes that the variance of the residuals is equal to the mean of the response variable. Consider the overdispersion statistic ϕ, which can be estimated using the following formula:

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^{n} \frac{(\text{art}_i - \lambda_i)^2}{\lambda_i},$$

where n is the sample size and p is the number of estimated coefficients in the model.

When $\phi > 1$ the response variable is overdispersed, which means that the Poisson model underestimates variability in the data, resulting in inflated test statistics (and deflated p-values).

(b) Use a simulation-based test for the hypothesis $H_0 : \phi = 1$ vs. $H_1 : \phi > 1$.

(c) To account for overdispersion, use a negative binomial regression instead of a Poisson regression to conduct a simulation-based test for the hypothesis $H_0 : b_1 = 0$ vs. $H_1 : b_1 \neq 0$.

(d) The negative binomial regression estimates a dispersion parameter θ. Obtain a bootstrapped distribution of $\hat{\theta}$ and compute its 95% BPI and a 95% CBCI.

(The following two exercises were not covered in the lectures yet, you are welcome to try them regardless. We will revisit the solutions after the content has been covered in lectures.)

(e) Obtain a confidence interval for b_1 based on model predictive simulations from the negative binomial regression model.

(f) Use the model implied log-likelihood as a loss metric. Calculate the loss for negative binomial regression using two models: One including predictor `ment` only, and one including predictors `phd` and `ment`. Obtain the in-sample loss, and an appropriate approximation of an out-of-sample loss. Compare the results and explain the pattern you found.